

Om Kumar Jha

LinkedIn: [linkedin.com/in/omkumarjha043](https://www.linkedin.com/in/omkumarjha043)
Github: github.com/PiUnknown

Email: okjha09@gmail.com
Mobile: +91-9311470171
Portfolio: piunknown.dev/

EDUCATION

Guru Gobind Singh Indraprastha University (GGSIPU)
Bachelor of Technology – Information Technology — GPA: 8

Delhi, India
Aug 2024 – May 2028

SKILLS SUMMARY

- **Languages:** Python, SQL, TypeScript
- **Frameworks:** FastAPI, Streamlit, React
- **ML / AI:** PyTorch, Scikit-learn, Numpy, Pandas, spaCy, LangChain, XGBoost, NVIDIA NIM, Groq API
- **RAG/LLMs:** ChromaDB, FAISS, Sentence-Transformers, HuggingFace Transformers
- **Cloud/DevOps:** AWS (EC2), Azure App Service, Docker, Git, GitHub Actions

EXPERIENCE

- **Indian Navy (WESEE)** Delhi, India
Data Science (Intern) Jul 2025 - Sep 2025
 - Built Python + SQL ingestion pipeline for high-frequency satellite data streams with interval-based sampling, reducing downstream load by 70–90% and stabilizing real-time analytics
 - Trained supervised ML models on geospatial data using cross-validation, improving detection consistency by 60–70%
 - Developed end-to-end ML workflows: preprocessing, feature engineering, hyperparameter tuning, and F1-score evaluation, reducing analysis time by ~40%
 - Designed risk-scoring algorithm using 5+ operational parameters, improving threat-zone prediction and reducing route-planning time by ~30%

PROJECTS

- **[Project Gnosis](#)** — Python, FastAPI, tree-sitter, NetworkX, ChromaDB, NVIDIA NIM, React:
 - Designed 7-agent pipeline that analyzes any GitHub repo and generates onboarding docs — architecture maps, dependency graphs, and tech debt reports
 - Implemented AST-based code chunking with tree-sitter (8 languages) for precise semantic retrieval, outperforming token-based chunking
 - Built NetworkX dependency graph with cycle detection and centrality analysis to identify critical files and hidden coupling
 - Deployed at gnosis.piunknown.dev
- **[TerraGraph](#)** — Python, FastAPI, ChromaDB, Sentence-Transformers, Groq API, Pydantic v2, Streamlit:
 - Built hybrid retrieval: deterministic graph matching (19 eco-relationships from FAO/IPCC/IPBES) + ChromaDB vector retrieval over 5 scientific PDFs
 - Enforced zero hallucination via Pydantic v2 schemas (7 required fields) with explicit constraints; <10 citation-backed recommendations per query
 - Implemented 5-stage routing: small-talk → on-topic → clarification → extraction → synthesis with unit-tested graph matching
 - Deployed at terragraph-x7oxucctkmfigypnt429be.streamlit.app
- **[IssueRouter](#)** — Python, FastAPI, BART, spaCy, Sentence-Transformers, Groq API, React:
 - Used all-MiniLM-L6-v2 to embed and cluster citizen complaints from X (Twitter) into unified issue groups, reducing redundant reports
 - Implemented LLM summarization via Groq API (llama3-8b-8192) for structured officer-ready action briefs from tweet clusters
 - Built end-to-end pipeline: tweet stream → zero-shot classification → embedding → clustering → summarization → dashboard

CERTIFICATIONS

[Supervised ML: Regression and Classification](#) | [RAG](#) | [AWS ML Foundations](#) | [Generative AI with AWS](#) | [Postman API Fundamentals](#) | [McKinsey Forward Scholar](#)

HONORS AND AWARDS

- **Harvard Project for Asian and International Relations (HPAIR)** 2025
 - Selected as a Delegate for HPAIR Asia Conference at The University of Tokyo through competitive global application
- **Trident Netra, Indian Navy (WESEE)** 2026
 - Contributed to Trident Netra, a naval AI system currently deployed and operational across Indian Navy vessels in real time
 - Project showcased at India AI Impact Summit 2026, Bharat Mandapam, New Delhi at the official Indian Navy stall